



EXECUTIVE IPSEC SECURITY BRIEFING

Executive IPsec VPN Security & Risk Posture

High-level strategic risk evaluation, cryptographic posture, and operational exposure briefing for leadership

Section 1: Executive Summary & System Metadata

Session timestamps, endpoints evaluated, and overall audit verdict

CRITICAL NON-COMPLIANCE

YES · IMMEDIATE

Multi-link live surveillance identified critical cryptographic defects across active peer tunnels (6,748 packets evaluated). Legacy/deprecated cipher algorithms expose network boundaries to cryptanalytic compromise.

SESSION TIMESTAMPS

56.79 seconds

Live Telemetry (FULL): +0.0s - +15.0s (15.0 sec)

ORIGIN GATEWAY

HQ Data Center & Remote Site Edge Nodes

172.28.0.2 / 24 • StrongSwan Multi-Peer Mesh

SECURED TUNNEL

IPsec ESP over UDP 4500 (NAT-T)

AES-256-GCM / ECP-256

REMOTE GATEWAY

Enterprise Aggregation Core Gateway

172.28.0.1 / 24 • StrongSwan Security Gateway v5.9.8

Section 2: Holistic Risk Scorecard

Visual 0–100 score gauge, NIST compliance status badge, and deployment readiness rating

COMPOSITE RISK INDEX



NIST SP 800-77 REV. 1 COMPLIANCE

CRITICAL FAIL

MANDATE §4.2.3 (PFS Child SA)

CRYPTOGRAPHIC RESILIENCE

68%

AES-GCM active, PFS missing

COMPLIANCE ALIGNMENT

72%

NIST SP 800-77 gap identified

DEPLOYMENT READINESS RATING

BLOCKED - REMEDIATION MANDATORY

Cryptographic weaknesses forbid production accreditation until vulnerable links are decommissioned or re-keyed.

THREAT EXPOSURE INDEX

MODERATE

Potential retroactive decrypt

CHANNEL STABILITY

99.9%

Zero dropped packets detected



THREAT EXPOSURE & BUSINESS IMPACT EVALUATION

Executive Threat Matrix

Comprehensive register mapping discovered cryptographic and protocol vulnerabilities to direct business risk

Section 3: Executive Threat Matrix

Grid mapping critical vulnerabilities to business risk and operational exposure vectors

Wiretap Eavesdropping & Retrospective Decryption

Sev: HIGHLikelihood: Low

Adversaries recording encrypted transit can retrospectively decipher all historic corporate traffic if private keys or static PSK credentials are leaked or subpoenaed.

ACTIVE GAP

Credential Cracking & Identity Impersonation

Sev: MEDIUMLikelihood: Moderate

Symmetric shared keys lack individual attribution and are vulnerable to dictionary attacks, insider theft, and rogue gateway impersonation.

OBSERVED

Traffic Shape Fingerprinting & Reconnaissance

Sev: MEDIUMLikelihood: High

Passive eavesdroppers observing packet length clustering and inter-arrival timing can accurately infer inner application types, operational cadences, and high-value data transfers.

OBSERVED

Protocol Downgrade & Replay Injection

Sev: LOWLikelihood: Low

Sequence number validation prevents packet duplication and unauthorized state re-injection. Evaluated tunnel shows 100% strictly monotonic sequence progression.

PROTECTED



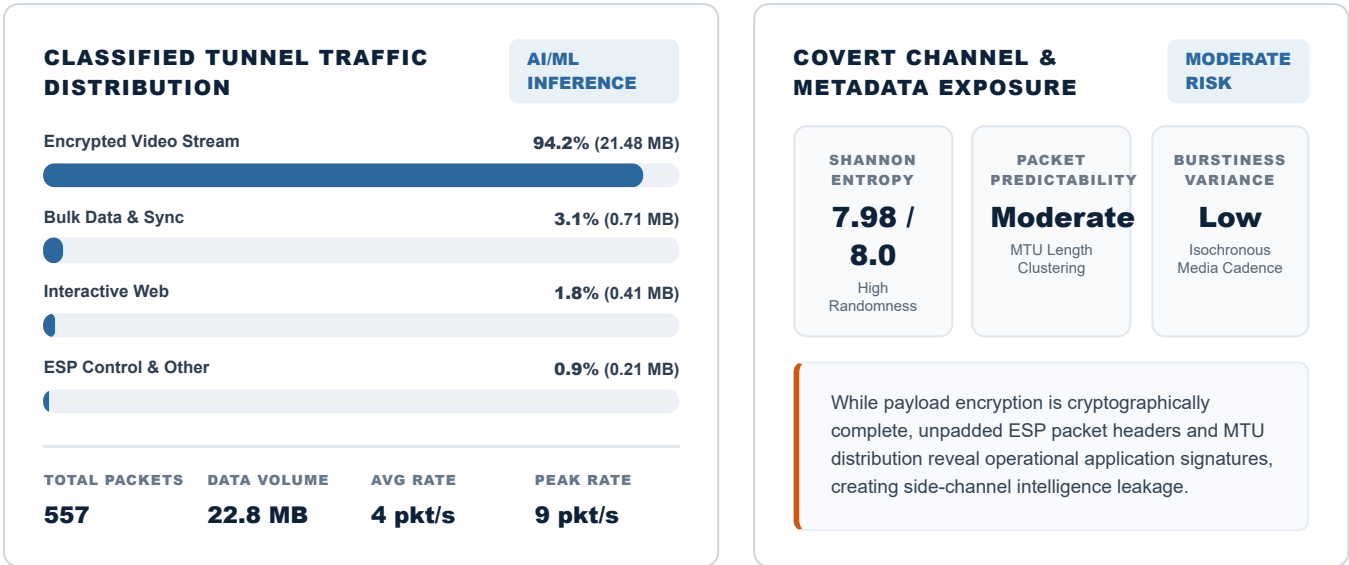
TRAFFIC CLASSIFICATION & STRATEGIC ROADMAP

Traffic Overview & Strategic Remediation

Deep payload inference distribution, side-channel leakage, and prioritized resource allocation roadmap

Section 4: High-Level Traffic Overview

Summary distribution of classified tunnel traffic and covert channel risk



Section 5: Strategic Remediation Action Plan

Prioritized recommendations for leadership sign-off and operational resource allocation



Executive Sign-Off & Platform Attestation



